HW-6

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question 1

(a) relation, primary key, and functional dependencies

an online game store keeps track of each users game library in one table:

`library(owner_id, game_id, username, game_title, developer, hours_played)`

Each row says that the user with `owner_id` (whose display name is `username`) owns the game `game_id` (with its `game_title` and `developer`) and has played it for `hours_played` hours.

primary key: `(owner_id, game_id)`. A user can own a given game only once, so the pair identifies a row, and `owner_id, game_id` $\rightarrow$ `username, game_title, developer, hours_played`.

non key functional dependencies:

FD1: `owner_id` $\rightarrow$ `username`. Each account has one display name, so every row for the same owner must show the same username.

FD2: `game_id` $\rightarrow$ `game_title, developer`. Each game in the store has one title and one developer, so every row for the same game must show the same title and developer.

Both are non trivial and non key dependencies

(b) instance with redundant data

owner_id	game_id	username	game_title	developer	hours_played
101	413150	grey	Stardew Valley	ConcernedApe	212
101	367520	grey	Hollow Knight	Team Cherry	48
202	413150	pixelpat	Stardew Valley	ConcernedApe	95
303	1145360	nova	Hades	Supergiant Games	30

The instance satisfies both FDs: owner 101 is always "grey", and game 413150 is always "Stardew Valley" by ConcernedApe. Its also redundant: "owner 101 is grey" is stored twice, and "413150 is Stardew Valley by ConcernedApe" is stored twice. With a real store, a popular game owned by millions of users would have its title and developer repeated millions of times.

(c) anomalies

update anomaly user 101 changes their username from "grey" to "greyson". If only the first row is updated, the second row still says "grey", so the table claims owner 101 has two different usernames. This breaks `owner_id` $\rightarrow$ `username`. To make the change correctly we have to update every row in user 101s library.

insertion anomaly The store adds a new game (say `game_id` 2000000, "Starlight Drift" by a new studio) before anyone has bought it. We cannot record its title and developer, because `owner_id` is part of the primary key and cannot be NULL, and there is no owner yet. The same problem happens for a new user who hasn't bought any games: their username can't be stored.

deletion anomaly user 303 refunds Hades, so their row is deleted. That row was the only one for user 303 and the only one for game 1145360, so deleting it also loses the fact that user 303 is "nova" and that Hades is made by Supergiant Games, even though we only meant to remove one purchase.

question 2

(a) decomposition: if $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$

step	FD	justification
1	$X \rightarrow YZ$	given
2	$YZ \rightarrow Y$	reflexivity, since $Y \subseteq YZ$
3	$X \rightarrow Y$	transitivity on (1) and (2)
4	$YZ \rightarrow Z$	reflexivity, since $Z \subseteq YZ$
5	$X \rightarrow Z$	transitivity on (1) and (4)

steps (3) and (5) give $X \rightarrow Y$ and $X \rightarrow Z$

(b) pseudotransitivity: if $X \rightarrow Y$ and $YW \rightarrow Z$, then $XW \rightarrow Z$

step	FD	justification
1	$X \rightarrow Y$	given
2	$XW \rightarrow YW$	augmentation of (1) with W
3	$YW \rightarrow Z$	given
4	$XW \rightarrow Z$	transitivity on (2) and (3)

step (4) gives $XW \rightarrow Z$

question 3

method. for a set of attributes X, write X^+ for its closure: every attribute that X determines. to find it, start with X (reflexivity) and keep applying FDs: if some FD $Y \rightarrow Z$ has Y already in the closure, add Z (this is augmentation/decomposition plus transitivity). X is a superkey exactly when X^+ contains every attribute of R, and a candidate key when no proper subset of X is also a superkey.

a useful shortcut: an attribute that never appears on the right-hand side of any FD can't be determined by anything else, so it has to be in *every* key.

(a) $R(a, b, c, d)$, FDs $\{b \to c,\ d \to a\}$

candidate key: bd (the only one).

- **determines everything:** $\{b, d\} \xrightarrow{b \to c} \{b, c, d\} \xrightarrow{d \to a} \{a, b, c, d\}$. so bd^+ is all of R.
- **minimal:** $b^+ = \{b, c\}$ (missing a, d) and $d^+ = \{a, d\}$ (missing b, c), so neither b nor d alone is a superkey.
- **no other keys:** b and d never appear on the right-hand side of an FD, so every superkey must contain both of them, i.e. every superkey contains bd. any superkey other than bd is then a proper superset of bd and isn't minimal.

(b) $R(a, b, c, d, e)$, FDs $\{a \to bc,\ cd \to e,\ b \to d,\ e \to a\}$

candidate keys: a, e, bc, cd.

here every attribute shows up on some right-hand side, so the shortcut doesn't help and we check closures directly.

- a: $\{a\} \xrightarrow{a \to bc} \{a, b, c\} \xrightarrow{b \to d} \{a, b, c, d\} \xrightarrow{cd \to e} \{a, b, c, d, e\}$. a single attribute is automatically minimal (its only proper subset is $\emptyset$, which determines nothing).
- e: $\{e\} \xrightarrow{e \to a} \{a, e\}$, and from a we get everything as above, so $e^+ = \{a, b, c, d, e\}$. minimal for the same reason.
- bc: $\{b, c\} \xrightarrow{b \to d} \{b, c, d\} \xrightarrow{cd \to e} \{b, c, d, e\} \xrightarrow{e \to a} \{a, b, c, d, e\}$. minimal since $b^+ = \{b, d\}$ and $c^+ = \{c\}$.
- cd: $\{c, d\} \xrightarrow{cd \to e} \{c, d, e\} \xrightarrow{e \to a} \{a, c, d, e\} \xrightarrow{a \to bc} \{a, b, c, d, e\}$. minimal since $c^+ = \{c\}$ and $d^+ = \{d\}$ (no FD has just c or just d on the left).

no other keys. take any superkey K.

- if K contains a or e, then K contains the key a or the key e, so it's only minimal if $K = a$ or $K = e$.
- otherwise $K \subseteq \{b, c, d\}$. the singletons fail ($b^+ = \{b, d\}$, $c^+ = \{c\}$, $d^+ = \{d\}$). of the pairs, bc and cd are keys, and $bd^+ = \{b, d\}$ is not a superkey. the only triple, bcd, contains bc, so it isn't minimal.

so the candidate keys are exactly a, e, bc, and cd.

(c) $R(a, b, c, d, e)$, FDs $\{c \to b,\ bd \to e,\ a \to d,\ e \to a\}$

candidate keys: ac, cd, ce.

c never appears on a right-hand side, so c is in every key. on its own, $c^+ = \{b, c\}$, which is not everything, so we need to add more.

- ac**:** $\{a, c\} \xrightarrow{c \to b} \{a, b, c\} \xrightarrow{a \to d} \{a, b, c, d\} \xrightarrow{bd \to e} \{a, b, c, d, e\}$. minimal since $a^+ = \{a, d\}$ and $c^+ = \{b, c\}$.
- cd**:** $\{c, d\} \xrightarrow{c \to b} \{b, c, d\} \xrightarrow{bd \to e} \{b, c, d, e\} \xrightarrow{e \to a} \{a, b, c, d, e\}$. minimal since $c^+ = \{b, c\}$ and $d^+ = \{d\}$.
- ce**:** $\{c, e\} \xrightarrow{c \to b} \{b, c, e\} \xrightarrow{e \to a} \{a, b, c, e\} \xrightarrow{a \to d} \{a, b, c, d, e\}$. minimal since $c^+ = \{b, c\}$ and $e^+ = \{a, d, e\}$.

no other keys. every key contains c. if a superkey K also contains a, d, or e, then it contains one of the keys ac, cd, or ce, so it's only minimal if it equals that key. otherwise $K \subseteq \{b, c\}$, but $c^+ = bc^+ = \{b, c\}$, so neither c nor bc is a superkey. so the candidate keys are exactly ac, cd, and ce.